# ASSIGNMENT-5

1. Explain the different types of the substations and draw layout for the same. Give the working of each component of a substation

2. Draw the layout for
   i) 11 kV substation
   ii) 33 kV substation
   iii) 66 kV substation
   iv) 220 kV substation

3. Calculate the charging current taken by an oil-filled cable of 7 km length having a core diameter 3.5 cm and inside sheath diameter of 7.8 cm. The cable is used as one phase of a 3-phase, 66 kV, 50 Hz system. Permitivity of the insulating material is 2.8.

4. The capacitance of a 3-core belted type cable are measured and found to be as follows.
   i) Between cores bunched together and sheath = 8 μF
   ii) Between conductor and the ~~two~~ other two connected together to sheath = 5 μF.

   Calculate the capacitance to neutral and the total charging kVA when the cable is connected to an 11 kV, 50 Hz, 3-phase system.

5. Estimate the charging current drawn by a cable with 3 cores and protected by a metal sheath when switched on to 11 kV, 3-$\phi$, 50 Hz supply. The capacitance measured between two cores with the third connected to the sheath is 3.7 μF.